



EDUFYI TECH SOLUTIONS

Virtual Internships · Industry Training · Placements

SPECIALISED PROFESSIONAL PROGRAM · 2026

HYBRID ELECTRIC VEHICLES

Powertrain architectures · Battery Management Systems · Motor control · Power electronics · Model-Based Design with MATLAB/Simulink · Functional safety.

6

Capstone
Projects

150+

Hiring
Partners

24x7

Mentor
Support

100%

Placement
Assistance

OUR ALUMNI WORK AT

Microsoft

BOSCH

Capgemini

Deloitte

IBM

Adobe

EY

TATA



Who we are, and why learners choose us

Edufyi Tech Solutions transforms the college-to-career journey with virtual internships and industry-grade training. Every learner ships six real-time and capstone projects under mentors working at MNCs and funded startups.

18.94%

Hybrid market CAGR

6

Industry projects

40+

Live mentor hours

BS6354

Wipro DICE ID

WHY CHOOSE US

01

Flexible pacing

Choose your speed, master the skills

02

Industry playground

Real OEM & Tier-1 style problem statements

03

Career grooming

Resume, aptitude, unlimited mocks & GDs

04

MNC mentors

Engineers from MNCs and startup founders

05

Placement support

100% assistance + 150+ hiring partners

06

R&D content

Exclusive research material from leading MNCs

07

Global certification

Wipro DICE ID accredited certificates

08

Partner badges

Microsoft, Cisco, IBM, Adobe, Autodesk

09

Lifetime LMS

Recorded lectures, anytime, anywhere

10

Paid performance

Performance based stipend opportunities

11

1:1 counselling

Personal roadmap and progress reviews

12

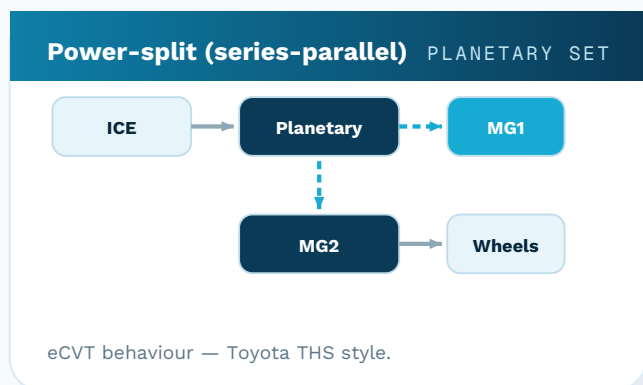
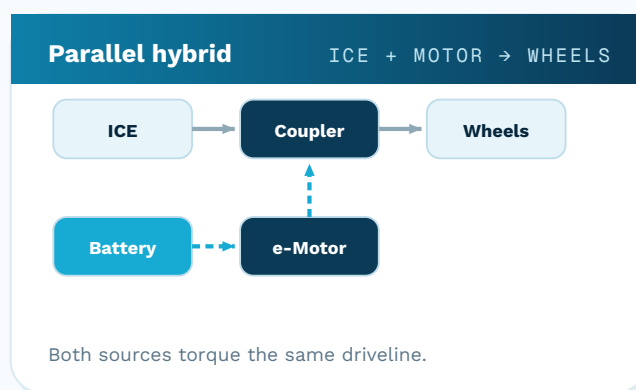
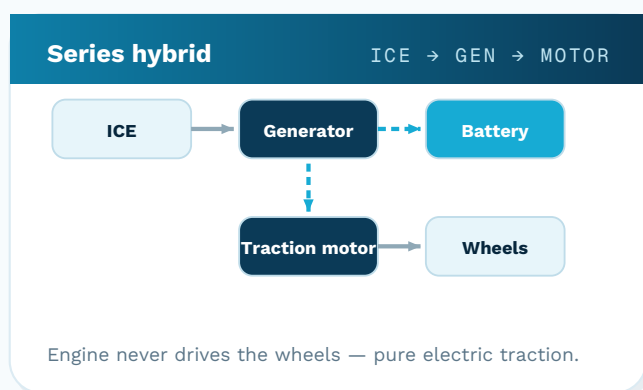
24x7 support

Dedicated doubt-resolution desk



What is a hybrid electric vehicle?

An HEV pairs an internal combustion engine with an electric propulsion system. Depending on the architecture, the electric path assists the engine, recovers braking energy, or drives the wheels on its own — improving efficiency, torque response and emissions.



DEGREE OF HYBRIDISATION

Micro	12V	Start-stop only
Mild	48V	Torque assist + regen
Full	200–400V	EV-only launch
PHEV	400V+	Grid chargeable, 40–80 km EV
REx	400V+	Range extender genset

HOW THE THREE POWERTRAINS COMPARE

PARAMETER	ICE	HEV	BEV
Tank-to-wheel efficiency	20–30%	35–45%	77–90%
Regenerative braking	None	Yes	Yes
Charging dependency	None	Optional	Essential
Tailpipe CO ₂	High	Reduced 25–40%	Zero
Powertrain complexity	Low	High (dual path)	Moderate

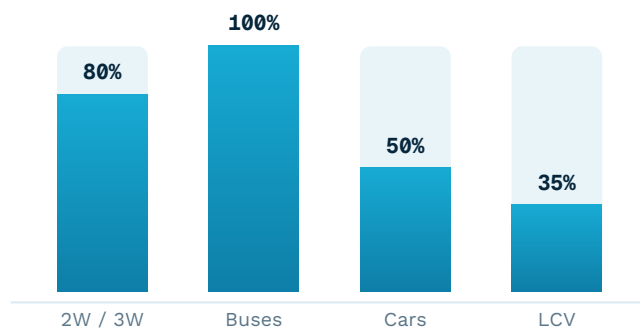


Scope of hybrid electric vehicles

Hybrids bridge the gap between combustion and full-electric mobility. With better cells, power electronics and software, they deliver efficiency today while charging infrastructure scales.

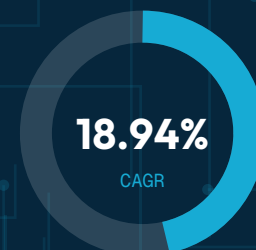
India electrification outlook - 2035

Share of new vehicle sales expected to be electrified



Hybrid market CAGR

2020 → 2030, by volume



Zero-emission national objectives keep funding and demand strong.

WHAT MAKES THIS PROGRAM ADVANCED

Model-Based Design

Simulink plant + controller, HIL-ready

Battery intelligence

SoC/SoH estimation with Kalman filters

Motor control

FOC for PMSM, space-vector PWM

Vehicle networks

CAN / CAN-FD, UDS diagnostics, AUTOSAR

Thermal & safety

Liquid cooling, ISO 26262, AIS-156

Energy management

Rule-based, ECMS and RL strategies

EMERGING TECHNOLOGY WATCHLIST - 2026

- 800 V architectures & SiC traction inverters
- Sodium-ion and silicon-anode chemistries
- Vehicle-to-grid (V2G) and smart charging
- OTA updates and software-defined vehicles
- Cell-to-pack and structural battery designs
- AI-driven BMS and predictive SoH analytics
- Digital twins for pack and drivetrain
- Hydrogen fuel-cell hybrid powertrains



Program curriculum · modules 01 – 05

Eight structured modules blend theory, hands-on labs and mentor-led design reviews. Each module closes with a graded assessment and a code or model deliverable.

01

HEV & EV Foundations

Live + LMS · 6 hrs

- Powertrain taxonomy: micro, mild, full, PHEV, REx, FCEV
- Vehicle dynamics: road load, rolling & aero resistance
- Otto / Atkinson cycles, BSFC maps and engine downsizing
- Regenerative braking energy budget and blending
- Well-to-wheel emissions and lifecycle CO₂ accounting

02

Energy Storage & Battery Systems

Live + Lab · 8 hrs

- Li-ion chemistries: NMC, LFP, LTO, and solid-state roadmap
- Cell modelling: OCV-R-RC equivalent circuits, parameter ID
- Pack design: series/parallel topology, busbars, fusing
- Ultracapacitors, flywheels and hybrid storage sizing
- Ageing, calendar/cycle life and second-life applications

03

Battery Management Systems (BMS)

Live + Project · 8 hrs

- SoC estimation: coulomb counting, EKF & UKF observers
- SoH / SoP estimation and impedance tracking
- Active vs passive cell balancing circuits
- Contactor control, precharge, insulation & HVIL monitoring
- Functional safety concept for a BMS ECU

04

Electric Machines & Motor Control

Live + Lab · 8 hrs

- PMSM, IPM, induction and SRM comparison for traction
- Clarke/Park transforms and Field Oriented Control
- Space-vector PWM, current loop tuning, dead-time effects
- Field weakening and MTPA operation for wide speed range
- Torque-speed envelope, efficiency maps, thermal derating

05

Power Electronics & Charging

Live + Lab · 8 hrs

- Traction inverters: IGBT vs SiC MOSFET trade-offs
- Bidirectional DC-DC converters and 48V architectures
- On-board charger, PFC stages and V2G / V2L power flow
- EVSE: AC Level 1/2, DC fast charging, CCS2 & GB/T
- ISO 15118 plug-and-charge, OCPP and smart charging



Program curriculum • modules 06 – 08

The second half moves from components to the complete vehicle: embedded software, model-based design, thermal safety and Indian policy landscape.

06

Embedded Software & Vehicle Networks

Live + Project • 8 hrs

- Automotive E/E architecture: domain vs zonal controllers
- CAN, CAN-FD, LIN, FlexRay and Automotive Ethernet
- UDS diagnostics, DTCs and secure flash bootloaders
- AUTOSAR Classic basics, RTE and MCAL layers
- Cybersecurity essentials: ISO/SAE 21434 overview

07

Model-Based Design & Simulation

Lab intensive • 10 hrs

- MATLAB / Simulink / Simscape plant modelling
- Drive-cycle simulation: MIDC, WLTP, FTP-75
- Energy management: rule-based, ECMS and DP benchmarks
- Auto-code generation, SIL / MIL / PIL and HIL testing
- Digital twin & OTA-ready calibration workflow

08

Thermal, Safety, Policy & Industry

Live • 6 hrs

- Battery & inverter thermal management, thermal runaway
- ISO 26262 ASIL decomposition and FMEA / FTA basics
- Homologation: AIS-038 Rev 2, AIS-156, BS-VI hybrids
- FAME-II, PM E-DRIVE, PLI schemes and demand incentives
- Charging infrastructure, battery swapping and business models

HOW EVERY MODULE IS DELIVERED

Live sessions

Instructor-led classes with recorded playback

Guided labs

Simulink, MATLAB and embedded C workbenches

Design reviews

Mentor critique on every model and report

Graded assessment

Quiz + deliverable scored against a rubric

ADVANCED ELECTIVES

- Reinforcement-learning energy management
- Fuel-cell hybrid powertrain modelling
- Battery ageing & second-life analytics
- HIL bench setup with dSPACE-style rigs



Six advanced industry projects

Every learner ships a portfolio of six graded projects, reviewed by mentors and packaged with documentation, test evidence and a demo video.

P1

48V mild-hybrid belt-starter-generator model

SIMULINK · DRIVE CYCLE

P2

EKF-based SoC estimator for an NMC pack

MATLAB · VALIDATION DATA

P3

FOC controller for a 60 kW traction PMSM

SIMULINK · SVPWM

P4

CAN-based BMS dashboard with UDS diagnostics

EMBEDDED C · CANOE-STYLE

P5

ECMS energy management for a power-split HEV

OPTIMISATION · DP BENCHMARK

P6

Fast-charging & thermal study of a 400V pack

THERMAL · CCS2 PROFILE

TOOLCHAIN YOU WILL MASTER

● MATLAB

● Simulink

● Simscape

● Stateflow

● Embedded C

● Python

● CANoe / CANalyzer

● Altium basics

● ANSYS Thermal

● GT-SUITE intro

● AUTOSAR

● Git

CAPSTONE SHOWCASE

Final demo day with hiring partners · portfolio + GitHub + technical viva

EVERY PROJECT SHIPS WITH

Requirement sheet

Scoped specification and acceptance criteria

Model / source code

Version-controlled Simulink model or C project

Test evidence

Drive-cycle results, plots and validation log

Technical report

Design rationale, trade-offs and conclusions

Demo video

Three-minute walkthrough for recruiters

Mentor score

Rubric grading with improvement feedback



Career opportunities & salary bands

Hybrid and EV programs at OEMs, Tier-1 suppliers and mobility startups hire across software, electrical and systems disciplines.

Role demand index & typical fresher-to-3yr band

EV Software Developer	4.5 - 12 LPA
System Engineer · Electric & Hybrid	5 - 14 LPA
Battery / BMS Engineer	4 - 11 LPA
Electric Motor Design Engineer	4 - 10 LPA
Power Electronics Engineer	5 - 13 LPA
Vehicle Integration Engineer	4.5 - 11 LPA
Model-Based Design Engineer	5 - 15 LPA
Charging Infrastructure Specialist	4 - 9 LPA

HOW TO START YOUR JOURNEY

01

Enroll

Pick your cohort & pace

02

Learn

Live + LMS modules

03

Build

6 mentor-reviewed projects

04

Get hired

Interviews with partners

WHERE OUR LEARNERS GET HIRED

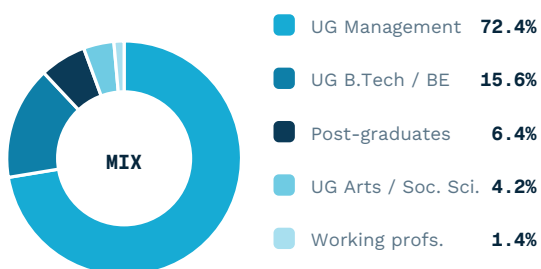
- **OEMs**
Tata Motors, Mahindra, Ashok Leyland class
- **Tier-1 suppliers**
Bosch, Valeo, Continental class
- **EV startups**
Battery, charging and drivetrain ventures
- **Engineering services**
R&D centres and MBD consultancies



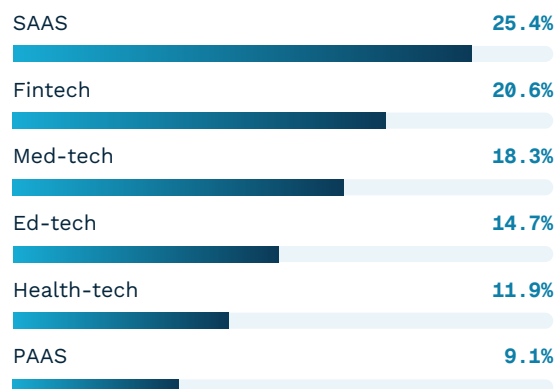
Placement & learner analytics

Transparent reporting on who learns with us, where they get placed and when offers land.

Learner background

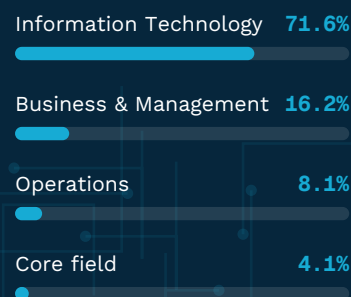
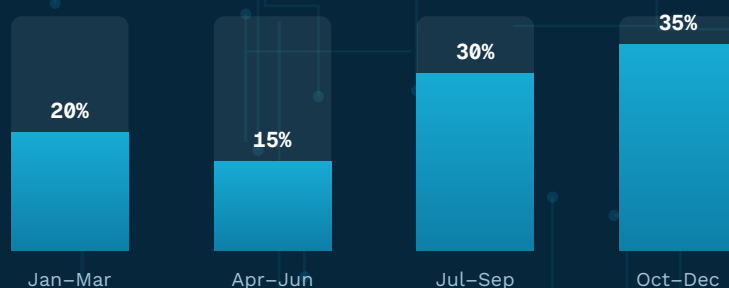


Placements by industry



Quarterly placement analysis

Share of annual offers released per quarter



CAREER SUPPORT INCLUDED

Resume & portfolio

ATS-ready CV plus a documented project portfolio

Mock interviews

Technical + HR rounds with domain mentors

Referral network

Introductions to hiring partners each cohort



Certifications you walk away with

Wipro DICE ID accredited certificates plus globally recognised professional badges from our authorised certification partners.

 EDUFYI TECH SOLUTIONS INTERNSHIP Completion Powered by Wipro DICE ID	 EDUFYI TECH SOLUTIONS PROJECT Completion Six graded capstone projects	 EDUFYI TECH SOLUTIONS INDUSTRIAL Training Domain specialisation record
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AUTHORISED CERTIFICATION PARTNERS

M Microsoft	C Cisco	I IBM	A Adobe
A Autodesk	I Intuit QuickBooks	E ESB	W Wipro DICE ID

ADDITIONAL ADVANTAGES

- Lifetime LMS access
- Recorded lectures
- Unlimited mocks & GDs
- Performance based stipend
- 1:1 counselling
- Access to hiring partners

PROGRAM AT A GLANCE

DURATION 6 months · weekend & weekday tracks	MODE Live online + recorded LMS + labs
ELIGIBILITY B.E / B.Tech / Diploma · Mech, EEE, ECE, Auto	ASSESSMENT Module quizzes, 6 graded projects, final viva
COHORT SIZE Capped for 1:1 mentor attention	SUPPORT Placement assistance until you are hired

ACCREDITED CERTIFICATIONS.

Every learner receives industry-recognised, accredited certificates on completion of the internship, industrial training and capstone project modules.



INTERNSHIP CERTIFICATE
Accredited Certification



INDUSTRIAL TRAINING
Accredited Certification



PROJECT COMPLETION
Accredited Certification

INTERNSHIP

Verified internship completion credential

TRAINING

Industrial training hours certified

PROJECT

Capstone project completion proof

GLOBAL MNC CERTIFICATIONS.

Learners are prepared for and certified through globally recognised professional certification programs from leading technology companies.



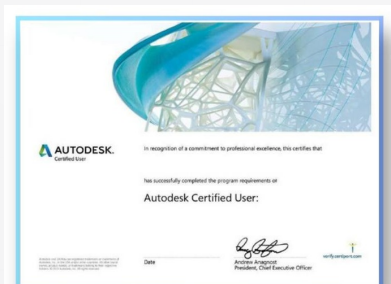
CISCO
Cybersecurity CCST



MICROSOFT
Azure Fundamentals



IBM
Internet of Things



AUTODESK
Certified User



ESB
Entrepreneurship

AUTHORISED PARTNERS.

Edufyi Tech Solutions is an authorised certification and testing partner for the world's leading technology certification bodies.



ADOBE



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QUICKBOOKS



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CISCO



ESB

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CERTIFY. BUILD. GET HIRED.



Accredited Certification



Microsoft Azure
Fundamentals



Cisco Cybersecurity CCST



IBM Internet of Things



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